

Parity Rock–Paper–Scissors Games and the Alper Constant

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Abstract

Two players choose numbers from $\{1, \dots, n\}$. Equal numbers draw, an odd difference is won by the smaller number, a nonzero even difference is won by the larger number, and a win with the number m pays $2m - h$, where $0 \leq h < 2$ is fixed. For $h = 1$ this is the game of Problem 1012 of Project Euler; for $h = 0$ the payment is proportional to the winning number. For every h outside an explicit countable set $\mathcal{E}$ we prove that the game has a unique equilibrium, supported on $2k + 1$ consecutive numbers ending at n or $n - 1$, and we solve it in closed form. When the support ends at n , the probability $P(n)$ of the largest number is $((2k^2 + 1)R_k - 2(k^2 - 1))/(3(2k + 1))$ for an explicit product R_k that depends only on the largest payment; otherwise $P(n) = 0$. The support widens at thresholds given by the zeros of these functions; for $h \in \{0, 1\}$ the thresholds are cubic polynomials in k for each parity. For $S_h(N) = \sum_{n=3}^N P(n)$ we prove $S_h(N) = (3/2)^{4/3} N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) + O_h(N^{-1/3})$, where $\tau \in [0, 1)$ is the relative position of N in its block, and we determine the next term. The constant $\mathcal{A}(h)$ plays the role that Euler’s constant plays for the harmonic series. We call $\mathcal{A} = \mathcal{A}(0) = 1.7771799879\dots$ the Alper constant, give a series and an integral representation, and certify 80 decimal places of $\mathcal{A}$ and of $\mathcal{A}(1) = 1.6464176534\dots$ with interval arithmetic. No expression of either constant in classical constants was found. The Alper function $h \mapsto \mathcal{A}(h)$ is strictly increasing and convex between consecutive points of $\mathcal{E}$ and jumps down at each of them; it tends to $\mathcal{A}(1)$ as $h \rightarrow 1^-$ and to $\mathcal{A}(0)$ as $h \rightarrow 2^-$, and $\mathcal{A}(h + 2) = \mathcal{A}(h)$ for $0 \leq h < 2$, $h \notin \mathcal{E}$, when the games are extended to larger parameters.

1 Introduction

In rock–paper–scissors every option beats exactly one other option. With equal payments the unique equilibrium is uniform. Unequal payments change the picture in a counterintuitive way. Suppose the loser pays 5 to a winning rock, 3 to a winning paper and 1 to winning scissors. Then the unique equilibrium plays rock, paper and scissors with probabilities $\frac{1}{9}$, $\frac{5}{9}$ and $\frac{1}{3}$: the most lucrative option is used least.

We study the following family of extensions to n options. Two players choose numbers from $\{1, \dots, n\}$ simultaneously. Equal numbers draw. If the numbers differ by an odd amount, the smaller number wins. If they differ by a nonzero even amount, the larger number wins. A win with the number m is paid $2m - h$ by the loser, where $0 \leq h < 2$ is a fixed parameter. We write $G_n(h)$ for this game. Every increasing payment schedule $c(m - \theta)$ with $c > 0$ and $0 \leq \theta < 1$ is covered after scaling, with $h = 2\theta$. Two members stand out:

- $h = 1$: the payments are 1, 3, 5, $\dots$. For $n = 3$ the numbers 1, 2, 3 play the roles of scissors, paper and rock with the payments 1, 3, 5 above. This game was posed as Problem 1012 of Project Euler [8], and a solution is described in [2].

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- $h = 0$: the payment is proportional to the winning number. This is the simplest member, and the one whose constant we call the Alper constant.

Whenever the equilibrium is unique, let $P(n)$ be the probability that it assigns to the largest number n , and put

$$S_h(N) = \sum_{n=3}^N P(n).$$

In particular, these quantities are unambiguous for $h \notin \mathcal{E}$, where the exceptional set $\mathcal{E}$ is defined below.

Solving $G_n(h)$ numerically for moderate n and generic h suggests that the equilibrium plays an odd number of consecutive numbers and that its width grows by two at isolated values of n . Sampled ratios $S_0(N)/N^{1/3}$ approach their limit slowly: the ratio is 1.330 at $N = 10^3$ and 1.613 at $N = 10^5$, while its limit turns out to be $(3/2)^{4/3} \approx 1.7171$. We prove the support and transition laws for every $h \notin \mathcal{E}$, and the asymptotic formulas first for $h \in \{0, 1\}$, where the thresholds are explicit polynomials, and then for every $h \notin \mathcal{E}$. They isolate the constant that governs the slow approach.

Throughout, $\log$ is the natural logarithm, γ is Euler's constant and $c_j = \binom{2j}{j} 4^{-j}$. For $k \geq 1$ and real y put

$$R_k(y) = \prod_{j=1}^k \frac{y - 2j}{y - 2j + 1}, \quad P_k(y) = \frac{(2k^2 + 1) R_k(y) - 2(k^2 - 1)}{3(2k + 1)}. \quad (1)$$

The variable y will be half of the largest payment in play, so $y = n - \frac{h}{2}$ when the largest number is n . For $k \geq 2$ the function P_k increases on $y > 2k$ and has exactly one zero β_k there (Lemma 5.2). Put $t_1(h) = 3$ and $t_k(h) = \lfloor \beta_k + \frac{h}{2} \rfloor + 1$ for $k \geq 2$: this is the first integer n with $P_k(n - \frac{h}{2}) > 0$. The integers n with $t_k(h) \leq n \leq t_{k+1}(h) - 1$ form the *block* k ; the blocks partition the integers $n \geq 3$. Finally let $\mathcal{E}$ be the set of $h \in [0, 2)$ for which $\beta_k + \frac{h}{2}$ is an integer for some $k \geq 2$. It is countable, and it contains neither 0 nor 1.

Theorem A (Theorems 4.3 and 5.4). *Let $0 \leq h < 2$, $h \notin \mathcal{E}$, and $n \geq 3$. The game $G_n(h)$ has a unique equilibrium. Let k be the block of n . If $n \leq t_{k+1}(h) - 2$, the support is $\{n - 2k, \dots, n\}$ and $P(n) = P_k(n - \frac{h}{2}) > 0$. If $n = t_{k+1}(h) - 1$, the support is $\{n - 1 - 2k, \dots, n - 1\}$ and $P(n) = 0$. The whole equilibrium is given in closed form by Theorem 4.3. For the two main cases*

$$t_k(0) = \left\lceil \frac{4k^3 + 5k}{6} \right\rceil + 1, \quad t_k(1) = \left\lfloor \frac{4k^3 + 5k}{6} \right\rfloor + 2.$$

The first values are $t_k(0) = 3, 8, 22, 47, 89, 150, 236, 349$ and $t_k(1) = 3, 9, 22, 48, 89, 151, 236, 350$.

Theorem B (Theorem 7.3). *Let $h \in \{0, 1\}$ and $\kappa = (3/2)^{4/3} = \sqrt[3]{81/16}$. For $N \geq 3$ let K be defined by $t_K(h) - 1 \leq N < t_{K+1}(h) - 1$, and put $\tau = (N - t_K(h) + 1)/(t_{K+1}(h) - t_K(h)) \in [0, 1)$. Then, uniformly in τ ,*

$$S_h(N) = \kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) - \frac{1 - 3\tau(1 - \tau)(4\tau - 1)}{4K} + O(N^{-2/3}).$$

Since $K = (3N/2)^{1/3} + O(1)$, the term with $1/K$ is of order $N^{-1/3}$. The quadratic term cannot be dropped: within every block, $S_h(N) - \kappa N^{1/3} + \frac{1}{4} \log N$ sweeps from about $-\mathcal{A}(h)$ down to about $-\mathcal{A}(h) - \frac{3}{8}$ and back. At the ends of the blocks, where P vanishes and $\tau = 0$, the oscillation disappears.

The Alper constant. The constant $\mathcal{A}(h)$ is an Euler-type constant. The harmonic numbers satisfy $H_N - \log N \rightarrow \gamma$; here, along the ends $N_K = t_{K+1}(h) - 1$ of the blocks,

$$\kappa N_K^{1/3} - \frac{1}{4} \log N_K - S_h(N_K) \rightarrow \mathcal{A}(h) \quad (K \rightarrow \infty).$$

We define the *Alper constant* as the constant of the simplest member, $h = 0$:

$$\mathcal{A} = \mathcal{A}(0) \approx 1.7771799879054231885280182162763415624255 \\ 7103207026437429288159344562194194456353$$

For the Project Euler game,

$$\mathcal{A}(1) \approx 1.6464176534405600338544841159858132863029 \\ 9683869991753007336025510008825020074857$$

Section 8 expresses both constants as rapidly computable series and as integral-series, and certifies that the values above are correctly rounded to 80 decimal places.

The Alper function. The limit above exists for every $h \notin \mathcal{E}$. This defines the *Alper function* $h \mapsto \mathcal{A}(h)$, whose value at 0 is the Alper constant (Section 10, Table 4 and Figure 2).

Theorem C (Theorem 10.3 and Propositions 10.4 and 10.5). *Let $h \in [0, 2) \setminus \mathcal{E}$, $\varphi = h - \lfloor h \rfloor$, and let K and τ be as in Theorem B. Then*

$$S_h(N) = \kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) - \frac{1 - 3\varphi - 3\tau(1 - \tau)(4\tau - 1)}{4K} + O_h(N^{-2/3}).$$

The function $\mathcal{A}$ is continuous, strictly increasing and strictly convex on every interval without points of $\mathcal{E}$, and it jumps down at every point of $\mathcal{E}$; at the first one, $6 - 2\sqrt{7}$, by $\frac{\sqrt{7}-1}{6}$. Moreover $\mathcal{A}(h) \rightarrow \mathcal{A}(1)$ as $h \rightarrow 1^-$, $\mathcal{A}(h) \rightarrow \mathcal{A}(0)$ as $h \rightarrow 2^-$, and $\mathcal{A}(h+2) = \mathcal{A}(h)$, where $\mathcal{A}(h+2)$ is defined by the same limit for the games $G_n(h+2)$.

The case $h = 1$ is a Project Euler problem, and its solvers may have obtained parts of our results for it, including the closed form and the first-order asymptotic formula; we make no priority claim for this case. The contributions developed here include the family $G_n(h)$, its transition law and asymptotic formulas for general h , the second-order term, the Alper constant with its certified digits, and the Alper function. Section 2 sets up the games, Section 3 shows that the supports are intervals, and Section 4 solves the equilibrium equations on an interval of odd length. Section 5 proves the transition law and Theorem A. Section 6 sums the probabilities over blocks. Section 7 proves Theorem B, Section 8 studies the constants, Section 9 compares the theorems with exact values of $S_0(10^m)$ for $m \leq 20$, and Section 10 proves Theorem C.

2 The games

Let a_{ij} be the payment received by a player who chooses i when the opponent chooses j . It is 0 for $i = j$. Otherwise the winning number w is $\min(i, j)$ when $i - j$ is odd and $\max(i, j)$ when $i - j$ is even, and the winner receives $2w - h$.

Lemma 2.1. *For all i, j , $a_{ij} = (i - j) + \operatorname{sgn}(i - j)(-1)^{i+j}(i + j - h)$. In particular $a_{ji} = -a_{ij}$.*

Proof. Take $j < i$. If $i - j$ is even, i wins and the formula gives $(i - j) + (i + j - h) = 2i - h$. If $i - j$ is odd, j wins and it gives $(i - j) - (i + j - h) = -(2j - h)$. The case $j > i$ follows by antisymmetry. $\square$

Against a vector $p = (p_1, \dots, p_n)$ the pure option i earns $f_i = \sum_j a_{ij}p_j$, and we use this formula for every integer i . Since $A = (a_{ij})$ is skew-symmetric, $p^\top A p = 0$, and the value of the game is 0. Hence a probability vector p is an equilibrium exactly when

$$p_i \geq 0, \quad \sum_i p_i = 1, \quad f_i \leq 0 \text{ for all } i, \quad f_i = 0 \text{ whenever } p_i > 0; \quad (2)$$

the last condition follows from the others because $\sum_i p_i f_i = p^\top A p = 0$. An equilibrium exists for every n by the minimax theorem [9]. The payoff rule does not depend on n , so the matrix of $G_{n-1}(h)$ is a principal submatrix of the matrix of $G_n(h)$.

For $n = 3$ the payoffs are $f_1 = (2 - h)p_2 - (6 - h)p_3$, $f_2 = -(2 - h)p_1 + (4 - h)p_3$ and $f_3 = (6 - h)p_1 - (4 - h)p_2$. Setting them to zero and normalising gives

$$p = \frac{(4 - h, 6 - h, 2 - h)}{12 - 3h},$$

so $p = (\frac{1}{3}, \frac{1}{2}, \frac{1}{6})$ for $h = 0$ and $p = (\frac{1}{3}, \frac{5}{9}, \frac{1}{9})$ for $h = 1$, the weighted rock–paper–scissors equilibrium of the introduction.

3 Interval supports

For a probability vector p let X_i be the probability that option i wins: the total mass of the numbers $j < i$ of the parity of i and of the numbers $j > i$ of the other parity.

Lemma 3.1. *For every probability vector p and every $i \leq n - 2$,*

$$f_{i+2} - f_i = 4(X_i - p_{i+1}) + (2i + 4 - h)(p_i + p_{i+2}) - (4i + 2 - 2h)p_{i+1}, \quad X_i \geq p_{i+1}.$$

Proof. The options i and $i + 2$ have the same parity, so they obtain the same result against every $j \notin \{i, i + 1, i + 2\}$. When both win, $i + 2$ receives 4 more. The three remaining columns give $a_{i+2,i} - a_{ii} = 2i + 4 - h$, $a_{i+2,i+1} - a_{i,i+1} = -(2i + 2 - h) - (2i - h) = -(4i + 2 - 2h)$ and $a_{i+2,i+2} - a_{i,i+2} = 2i + 4 - h$. Option i beats $i + 1$, which gives $X_i \geq p_{i+1}$. $\square$

Proposition 3.2. *Let $n \geq 3$ and let p be an equilibrium of $G_n(h)$. Its support is an interval $\{a, \dots, b\}$ with $b \in \{n - 1, n\}$ and $b > a$.*

Proof. Suppose $p_i > 0$ and $p_{i+1} = 0$ for some $i \leq n - 2$. Then $f_i = 0$ and Lemma 3.1 gives $f_{i+2} \geq (2i + 4 - h)p_i > 0$, which contradicts (2). So every chosen number below $n - 1$ is followed by another chosen number, and the support is an interval $\{a, \dots, b\}$ with $b \geq n - 1 \geq 2$. A single number $b \geq 2$ is not an equilibrium, because $b - 1$ beats it and receives $2b - 2 - h > 0$. $\square$

For $h \notin \mathcal{E}$ the support has odd size, as proved in Proposition 5.3. For the payments $1, 3, 5, \dots$ of $h = 1$ there is a short argument modulo 2, but for general nonexceptional h the parity of the support follows from the explicit solution of the next section.

4 The equilibrium on an interval of odd length

Fix $k \geq 1$, $b \geq 2k + 1$, $a = b - 2k$, $L = 2k + 1$, $I = \{a, \dots, b\}$ and $y = b - \frac{h}{2}$, so that $2y$ is the payment for a win with b . For a real vector p supported on I put

$$s = \sum_j p_j, \quad M = \sum_j j p_j, \quad q_i = \sum_{j \leq i} (-1)^{i-j} p_j \quad (i \in \mathbb{Z}), \quad Q = q_b.$$

Thus $q_i = 0$ for $i < a$, $q_i = (-1)^{i-b} Q$ for $i \geq b$, and $p_i = q_i + q_{i-1}$. The numbers of the parity of b carry the mass $\frac{s+Q}{2}$, and the others carry $\frac{s-Q}{2}$.

Lemma 4.1 (moment equation). *For $a \leq j \leq b$,*

$$\sum_{i=a}^b a_{ij} = L\left(\frac{a+b}{2} - j\right) + \begin{cases} k, & j \equiv b \pmod{2}, \\ -(k+1), & j \not\equiv b \pmod{2}, \end{cases}$$

independently of h . Consequently, if $f_i = 0$ for all $i \in I$, then $L(\frac{a+b}{2}s - M) + k\frac{s+Q}{2} - (k+1)\frac{s-Q}{2} = 0$; for $s = 1$ this reads $M - \frac{Q}{2} = \frac{a+b}{2} - \frac{1}{2L}$, and for $s = 0$ it reads $M = \frac{Q}{2}$.

Proof. The column sum follows from Lemma 2.1 by pairing consecutive terms; the terms containing h cancel. Adding the L equations $f_i = 0$ gives the stated equation. $\square$

Lemma 4.2 (neighbour identity). *For every real vector p supported on I and every integer i ,*

$$f_i + f_{i+1} = (2i+1)s - 2M + (-1)^{b-i}Q + (2i+2-h)q_{i+1} - (2i-h)q_{i-1}.$$

Proof. With $\varepsilon_j = (-1)^j$, Lemma 2.1 gives $f_i = is - M + \varepsilon_i[(i-h)T_i + U_i]$, where $T_i = \sum_{j < i} \varepsilon_j p_j - \sum_{j > i} \varepsilon_j p_j$ and $U_i = \sum_{j < i} \varepsilon_j j p_j - \sum_{j > i} \varepsilon_j j p_j$. Passing from i to $i+1$ changes T_i and U_i only in the terms of p_i and p_{i+1} , and $\varepsilon_{i+1} = -\varepsilon_i$, so $f_i + f_{i+1} = (2i+1)s - 2M - \varepsilon_i T_i - (2i+1-h)p_i + (2i+2-h)p_{i+1}$. Finally $\varepsilon_i T_i = q_i - q_{i-1} - (-1)^{b-i}Q$, and $p_i = q_i + q_{i-1}$, $p_{i+1} = q_{i+1} + q_i$. $\square$

If $s = 1$ and $f_i = 0$ on I , the identity for $a \leq i \leq b-1$ gives the $2k$ neighbour equations

$$(2i+2-h)q_{i+1} - (2i-h)q_{i-1} = 2M - (2i+1) - (-1)^{b-i}Q \quad (a \leq i \leq b-1). \quad (3)$$

Each equation links q_{i-1} and q_{i+1} . The unknowns therefore split into the chain $a-1, a+1, \dots, b-1$ and the chain $a, a+2, \dots, b$.

Theorem 4.3. *Let p be a real vector supported on I with $\sum_j p_j = 1$ and $f_i = 0$ for all $i \in I$. Put $R = R_k(y)$, $c_h = \frac{1-h}{3}$ and*

$$K = \frac{a+b-1}{2} - \frac{1}{2L}, \quad Q = \frac{2y(R-1) + 2k}{3} + \frac{1}{L}.$$

Then $p_m = q_m + q_{m-1}$ with

$$q_m = K + c_h - \frac{m}{3} - \left(K + c_h - \frac{a-1}{3}\right) \prod_{\substack{a \leq i \leq m-1 \\ i \equiv b \pmod{2}}} \frac{2i-h}{2i+2-h} \quad (m = a-1, a+1, \dots, b-1), \quad (4)$$

$$q_m = K + c_h + Q - \frac{m}{3} + \left(\frac{b}{3} - K - c_h\right) \prod_{\substack{m+1 \leq i \leq b-1 \\ i \not\equiv b \pmod{2}}} \frac{2i+2-h}{2i-h} \quad (m = a, a+2, \dots, b). \quad (5)$$

In particular

$$p_b = P_k(y), \quad p_a = K + c_h + Q - \frac{a}{3} + \left(\frac{b}{3} - K - c_h\right) \frac{yR}{y-2k}.$$

Proof. On the first chain the equations (3) have $(-1)^{b-i} = 1$. The homogeneous step multiplies by $\frac{2i-h}{2i+2-h}$. A function $\gamma' - \frac{m}{3}$ is a particular solution exactly when $2\gamma' = 2M - 1 - Q + \frac{2-2h}{3}$, because $(2i+2-h)(i+1) - (2i-h)(i-1) = 6i+2-2h$. This gives $\gamma' = M - \frac{1+Q}{2} + c_h = K + c_h$, by Lemma 4.1. Starting from $q_{a-1} = 0$ gives (4). Over the whole chain the product is $R_k(y)$.

On the second chain $(-1)^{b-i} = -1$, the particular solution is $K + c_h + Q - \frac{m}{3}$, and the chain ends at $q_b = Q$. Running it downwards gives (5). The full product over the second chain is $1/R_k(y+1)$, and the telescoping identity

$$R_k(y) R_k(y+1) = \frac{y-2k}{y} \quad (6)$$

gives the formula for $p_a = q_a$.

It remains to determine Q . Add the equations (3) for $a \leq i \leq b-1$:

- the left sides telescope to $(2b - h)Q + (2b - 2 - h)q_{b-1} - (2a + 2 - h)q_a - 2 \sum_{m=a+1}^{b-2} q_m$;
- the terms $\pm Q$ on the right cancel, and $\sum_{i=a}^{b-1} (2i + 1) = b^2 - a^2$.

The normalisation $1 = Q + 2 \sum_{m=a}^{b-1} q_m$ removes the remaining sum. Together with Lemma 4.1 this yields

$$2y p_b - (2a - h) p_a = (2k - 1) Q + \frac{1}{L}, \quad p_b = Q + q_{b-1}. \quad (7)$$

Inserting the two chain solutions leaves a linear equation for Q in which Q has the coefficient L . Its solution is the stated value. Then $p_b = Q + q_{b-1} = P_k(y)$. $\square$

Lemma 4.4 (the kernel). *For every $0 \leq h < 2$ the kernel of the principal submatrix A_I is a line, spanned by a vector with nonzero coordinate sum. Consequently the formulas of Theorem 4.3 define a vector with $\sum_j p_j = 1$ and $f_i = 0$ on I , and it is the only such vector.*

Proof. A skew-symmetric matrix of odd order is singular, so A_I has a nonzero kernel vector w . Suppose that $\sum_j w_j = 0$. Lemma 4.1 gives $M = \frac{Q}{2}$, and Lemma 4.2 turns the equations $f_i + f_{i+1} = 0$, $a \leq i \leq b - 1$, into $(2i + 2 - h)q_{i+1} - (2i - h)q_{i-1} = 2M - (-1)^{b-i}Q$. On the first chain the right side is 0, so $q \equiv 0$ there, because $q_{a-1} = 0$. On the second chain the right side is $2Q$; the constant Q is a solution, and since $q_b = Q$ the solution is this constant. Hence $w_m = q_m + q_{m-1} = Q$ for all $m \in I$, so $0 = \sum_j w_j = LQ$ and $w = 0$, a contradiction. Thus every nonzero kernel vector has nonzero sum and can be normalised; Theorem 4.3 shows that the normalised kernel vector is unique, so the kernel is a line. $\square$

Example 4.5. Take $h = 0$, $b = 10$ and $k = 2$, so $I = \{6, \dots, 10\}$, $y = 10$ and $R_2(10) = \frac{8}{9} \cdot \frac{6}{7} = \frac{16}{21}$. The theorem gives

$$(p_6, p_7, p_8, p_9, p_{10}) = \left(\frac{29}{315}, \frac{92}{315}, \frac{34}{105}, \frac{74}{315}, \frac{2}{35} \right),$$

with $Q = -\frac{17}{315}$ and $M = \frac{496}{63}$, so that $M - \frac{Q}{2} = \frac{79}{10} = \frac{6+10}{2} - \frac{1}{10}$. This is the equilibrium of $G_{10}(0)$, and $P(10) = \frac{2}{35}$.

5 Neighbouring options and the transition law

The interval solution decides its own fate through the two options next to the support.

Proposition 5.1. *In the setting of Theorem 4.3,*

$$f_{a-1} = \frac{2k+3}{2k+1} 2y R_k(y) P_{k+1}(y+1), \quad f_{b+1} = -\frac{2k+3}{2k+1} (2y+2) P_{k+1}(y+2).$$

Proof. Lemma 4.2 with $i = a - 1$ (where $f_a = 0$, $q_{a-2} = 0$ and $(-1)^{b-a+1} = -1$) and with $i = b$ (where $f_b = 0$, $q_{b+1} = -Q$ and $q_{b-1} = p_b - Q$) gives

$$f_{a-1} = (2a - 1) - 2M - Q + (2a - h) p_a, \quad f_{b+1} = (2b + 1) - 2M - Q - 2y p_b.$$

By Lemma 4.1, $2M + Q = a + b - \frac{1}{L} + 2Q$, and by (7),

$$f_{a-1} = 2y p_b - L(1 + Q), \quad f_{b+1} = L + \frac{1}{L} - 2Q - 2y p_b.$$

Inserting $p_b = P_k(y)$ and the value of Q from Theorem 4.3, both payoffs become affine in $R = R_k(y)$:

$$\begin{aligned} 3L f_{a-1} &= (2k^2 + 4k + 3)(2y - 4k - 2) - 2k(k + 2) 2y R, \\ 3L f_{b+1} &= 2k(k + 2)(2y + 2) - (2k^2 + 4k + 3) 2y R. \end{aligned}$$

On the other hand, the definition of R_{k+1} gives

$$R_{k+1}(y+1)R_k(y) = \frac{y-2k-1}{y}, \quad R_{k+1}(y+2) = \frac{y}{y+1}R_k(y).$$

Since $2(k+1)^2 + 1 = 2k^2 + 4k + 3$ and $2((k+1)^2 - 1) = 2k(k+2)$, the definition (1) of P_{k+1} turns the right-hand sides into $3(2k+3)2yR_{k+1}(y+1)$ and $-3(2k+3)(2y+2)P_{k+1}(y+2)$. $\square$

Lemma 5.2 (root bound). *Let $k \geq 2$. On the half-line $y > 2k$ the function P_k is continuous and strictly increasing, and its unique zero β_k satisfies*

$$\frac{4k^3 + 5k + 3}{6} < \beta_k < \frac{4k^3 + 5k + 6}{6}.$$

In particular $\beta_k \notin \frac{1}{2}\mathbb{Z}$. For $k = 1$ we have $P_1(y) = \frac{y-2}{3(y-1)} > 0$ for $y > 2$.

Proof. For $y > 2k$ every factor of $R_k(y)$ is positive and increasing, so R_k increases strictly from 0 to 1, and $P_k(y) = 0$ exactly when $-\log R_k(y) = \lambda_k := \log \frac{2k^2+1}{2k^2-2}$. Put $x = y - k - \frac{1}{2}$ and

$$U = \bigcup_{j=1}^k [k-2j + \frac{1}{2}, k-2j + \frac{3}{2}], \quad G(x) = \int_U \frac{dt}{x+t}.$$

The interval with index j contributes $\log \frac{y-2j+1}{y-2j}$, so $G(x) = -\log R_k(y)$. The set U is symmetric about 0, has measure k and lies in $[-d, d]$ with $d = k - \frac{1}{2}$; on $x > d$ the function G is strictly decreasing. Let x^* be the solution of $G(x^*) = \lambda_k$, so that $\beta_k = x^* + k + \frac{1}{2}$.

Lower bound. Let $x_0 = k(4k^2 - 1)/6 > d$. Then $x_0 + \frac{k}{2} = \frac{k(2k^2+1)}{3}$ and $x_0 - \frac{k}{2} = \frac{k(2k^2-2)}{3}$, so $\int_{-k/2}^{k/2} \frac{dt}{x_0+t} = \lambda_k$. Folding both integrals onto $t \geq 0$ gives

$$G(x_0) - \lambda_k = \int_{U_+} g(t) dt - \int_0^{k/2} g(t) dt, \quad g(t) = \frac{2x_0}{x_0^2 - t^2}, \quad U_+ = U \cap [0, \infty).$$

The set U_+ has measure $\frac{k}{2}$, and since $k \geq 2$ it has positive measure beyond $\frac{k}{2}$. As g is strictly increasing on $[0, x_0)$, the difference is positive. Hence $G(x_0) > \lambda_k$ and $x^* > x_0$.

Upper bound. Let $z = x_0 + \frac{1}{2}$. By symmetry $G(z) = \int_U \frac{z dt}{z^2 - t^2}$, and for $|t| \leq d$ we have $\frac{z}{z^2 - t^2} \leq \frac{1}{z}(1 + \frac{t^2}{z^2 - d^2})$. Since $\int_U t^2 dt = kv$ with $v = \frac{4k^2-3}{12}$,

$$G(z) \leq \frac{k}{z} \left(1 + \frac{v}{z^2 - d^2}\right) < \frac{k}{x_0}.$$

The second inequality is equivalent to $2x_0v < z^2 - d^2 = x_0^2 + x_0 - (k^2 - k)$, which follows from $x_0 > 2v$ and $x_0 > k^2 - k$. Finally $\frac{k}{x_0} = \frac{6}{4k^2-1} = \frac{2u}{2+u}$ with $u = \frac{3}{2k^2-2}$, and $\frac{2u}{2+u} < \log(1+u) = \lambda_k$. Hence $G(z) < \lambda_k$ and $x^* < z$.

Thus $x_0 < x^* < x_0 + \frac{1}{2}$, which is the stated bound because $x_0 + k + \frac{1}{2} = \frac{4k^3+5k+3}{6}$. The number $\frac{4k^3+5k}{6}$ is an integer for even k and a half-integer for odd k , so the two bounds are consecutive points of $\frac{1}{2}\mathbb{Z}$. $\square$

By Lemma 5.2, $t_k(h) = \lfloor \beta_k + \frac{h}{2} \rfloor + 1$ is the first integer n with $P_k(n - \frac{h}{2}) > 0$, and for $h \notin \mathcal{E}$ no integer n gives $P_k(n - \frac{h}{2}) = 0$. Since $\beta_k \notin \frac{1}{2}\mathbb{Z}$, neither 0 nor 1 lies in $\mathcal{E}$, and the bounds of Lemma 5.2 give the formulas for $t_k(0)$ and $t_k(1)$ stated in Theorem A, equivalently

$$t_k(h) = \frac{2}{3}k^3 + \frac{5}{6}k + \frac{5+2h-(-1)^{k+h}}{4} \quad (h \in \{0, 1\}). \quad (8)$$

Proposition 5.3. *Let $h \notin \mathcal{E}$. Every equilibrium of $G_n(h)$ has a support of odd size $2k + 1 \geq 3$.*

Proof. Suppose the support $I_0 = \{a, \dots, b\}$ of an equilibrium p has even size $2k \geq 2$, and let $I = \{a, \dots, b + 1\}$; the option $b + 1$ may lie outside the game, and f_{b+1} is computed with Lemma 2.1. On I we have $A_I p = \varphi e_{b+1}$ with $\varphi = f_{b+1}$, because $f_i = 0$ on I_0 . Let v be the normalised kernel vector of A_I from Lemma 4.4. By Theorem 4.3, $v_{b+1} = P_k(b + 1 - \frac{h}{2})$, which is nonzero: for $k \geq 2$ because $h \notin \mathcal{E}$, and for $k = 1$ by Lemma 5.2. Skew-symmetry gives $0 = -(A_I v)^\top p = v^\top A_I p = \varphi v_{b+1}$, so $\varphi = 0$. Then p is a normalised kernel vector of A_I , so $p = v$ and $p_{b+1} = v_{b+1} \neq 0$, a contradiction. $\square$

Theorem 5.4. *Let $h \notin \mathcal{E}$ and let $n \geq 3$ lie in the block k . The game $G_n(h)$ has exactly one equilibrium. If $n \leq t_{k+1}(h) - 2$, its support is $\{n - 2k, \dots, n\}$ and $P(n) = P_k(n - \frac{h}{2}) > 0$. If $n = t_{k+1}(h) - 1$, its support is $\{n - 1 - 2k, \dots, n - 1\}$ and $P(n) = 0$.*

Proof. Write t_k for $t_k(h)$. Let p be an equilibrium of $G_n(h)$; it exists by the minimax theorem. By Propositions 3.2 and 5.3 its support is $\{a, \dots, b\}$ with $b \in \{n - 1, n\}$ and $b - a = 2k'$ for some $k' \geq 1$, and by Theorem 4.3, $p_b = P_{k'}(b - \frac{h}{2})$.

First, $t_{k'} \leq b \leq t_{k'+1} - 2$. Indeed $P_{k'}(b - \frac{h}{2}) = p_b > 0$, so $b \geq t_{k'}$. If $a \geq 2$, the option $a - 1$ exists and $f_{a-1} \leq 0$. As $R_{k'}(b - \frac{h}{2}) > 0$, Proposition 5.1 gives $P_{k'+1}(b + 1 - \frac{h}{2}) \leq 0$, hence < 0 , and so $b + 1 < t_{k'+1}$. If $a = 1$, then $b = 2k' + 1 \leq t_{k'+1} - 2$, because $t_{k'+1} > \beta_{k'+1} > 2k' + 3$ by Lemma 5.2.

If $b = n - 1$, the option $n = b + 1$ exists and $f_n \leq 0$, so $P_{k'+1}(b + 2 - \frac{h}{2}) \geq 0$ by Proposition 5.1, hence > 0 , and $b + 2 \geq t_{k'+1}$. Together with the first step, $n = t_{k'+1} - 1$; this is the last element of the block k' , so $k' = k$, $n = t_{k+1} - 1$ and $P(n) = p_n = 0$.

If $b = n$, the first step gives $t_{k'} \leq n \leq t_{k'+1} - 2$, so $k' = k$, $n \leq t_{k+1} - 2$ and $P(n) = p_n = P_k(n - \frac{h}{2}) > 0$.

The two cases are distinguished by whether $n = t_{k+1} - 1$, so the support is determined by n . By Lemma 4.4, p is the only normalised vector on this support with vanishing payoffs on it. Hence the equilibrium is unique. $\square$

Remark 5.5. The proof never checks that the closed-form probabilities are positive or that the excluded options lose. Positivity and the sign conditions are inherited from the equilibrium whose existence the minimax theorem guarantees; as a by-product, every p_m in Theorem 4.3 is positive when $t_k(h) \leq b \leq t_{k+1}(h) - 2$.

Write $\beta_k = \frac{4k^3 + 5k + 3}{6} + \vartheta_k$ with $0 < \vartheta_k < \frac{1}{2}$. Unwinding the definition of $t_k(h)$,

$$\mathcal{E} = \{1 - 2\vartheta_k : k \geq 2 \text{ even}\} \cup \{2 - 2\vartheta_k : k \geq 3 \text{ odd}\}.$$

For even k , $t_k(h)$ equals $t_k(0)$ when $h < 1 - 2\vartheta_k$ and $t_k(1)$ when $h > 1 - 2\vartheta_k$. For odd k , $t_k(0) = t_k(1)$, and $t_k(h)$ equals this common value when $h < 2 - 2\vartheta_k$ and exceeds it by one when $h > 2 - 2\vartheta_k$. In fact,

$$\vartheta_k = \frac{3}{8k} + O(k^{-3}).$$

To prove this, retain U , G and $x_0 = k(4k^2 - 1)/6$ from the proof of Lemma 5.2, and put $V = [-k/2, k/2]$. Then $\lambda_k = \int_V (x_0 + t)^{-1} dt$. Both sets are symmetric, have measure k and lie in $[-k, k]$. Expanding the integrands, with the odd moments cancelling, gives

$$G(x_0) - \lambda_k = \frac{\int_U t^2 dt - \int_V t^2 dt}{x_0^3} + O\left(\frac{k^5}{x_0^5}\right) = \frac{k(k^2 - 1)}{4x_0^3} + O\left(\frac{k^5}{x_0^5}\right).$$

Here $\int_U t^2 dt = k(4k^2 - 3)/12$ and $\int_V t^2 dt = k^3/12$. Uniformly for $x_0 \leq x \leq x_0 + 1/2$, symmetry also gives $-G'(x) = kx_0^{-2}(1 + O(k^{-3}))$. Since $G(x_0 + \vartheta_k) = \lambda_k$ and $0 < \vartheta_k < 1/2$, the mean value theorem yields

$$\vartheta_k = \frac{k^2 - 1}{4x_0} (1 + O(k^{-3})) + O\left(\frac{k^4}{x_0^3}\right) = \frac{3}{8k} + O(k^{-3}).$$

Consequently $\mathcal{E}$ accumulates only at 1 and 2, and for every fixed $h \notin \mathcal{E}$ the thresholds eventually agree with those of $h = 0$ (if $h < 1$) or of $h = 1$ (if $h \geq 1$). For illustration, $\vartheta_2 \approx 0.1458$, $\vartheta_{10} \approx 0.0372$ and $\vartheta_{400} \approx 0.00094$; monotonicity is not needed.

For $h \in \mathcal{E}$ uniqueness fails for at least one game: if $\beta_k + \frac{h}{2} = m \in \mathbb{Z}$, let nonexceptional parameters tend to h from both sides in Theorem 5.4. The limiting vectors are supported on $\{m - 2k + 1, \dots, m - 1\}$ and $\{m - 2k, \dots, m - 2\}$. They are equilibria of $G_{m-1}(h)$ because the equilibrium inequalities are continuous. They are distinct: the first assigns probability $P_{k-1}(\beta_k - 1) > 0$ to $m - 1$, while the second assigns zero. The strict inequality follows from Lemma 5.2, which gives $\beta_k - 1 > \beta_{k-1}$ for $k \geq 3$; for $k = 2$ use $P_1(y) > 0$ for $y > 2$. Their convex combinations are also equilibria.

6 Block sums

In Sections 6–9, $h \in \{0, 1\}$; general h is treated in Section 10. By Theorem 5.4, S_h is a sum over blocks. For $k \geq 1$ put

$$\ell_k = t_{k+1}(h) - t_k(h) - 1 = 2k^2 + 2k + \frac{1 + (-1)^{k+h}}{2}, \quad B_k = B_k(h) = \sum_{n=t_k(h)}^{t_{k+1}(h)-2} P_k\left(n - \frac{h}{2}\right),$$

so that $S_h(t_{K+1}(h) - 1) = \sum_{k=1}^K B_k$. Table 1 lists the first blocks for $h = 0$.

k	$t_k(0)$	n with $P(n) = P_k(n) > 0$	zero of P	ℓ_k	$B_k(0)$
1	3	3–6	7	4	0.9055555556
2	8	8–20	21	13	1.2051685518
3	22	22–45	46	24	1.2449235895
4	47	47–87	88	41	1.3328846236
5	89	89–148	149	60	1.3441658999
6	150	150–234	235	85	1.3842642085

Table 1: The first blocks of the game $G_n(0)$. In the block k the support has $2k + 1$ elements; at the zero of P it is shifted down by one.

Lemma 6.1 (partial fractions). *For $k \geq 1$,*

$$1 - R_k(y) = \sum_{j=1}^k \frac{w_{k,j}/2}{y - 2j + 1}, \quad w_{k,j} = 2(2k - 2j + 1) c_{j-1} c_{k-j}, \quad \sum_{j=1}^k w_{k,j} = 2k.$$

Moreover $w_{k,j}/(2k) = \Pr(X = j - 1)$, where X has the beta-binomial distribution with parameters $k - 1$, $\frac{1}{2}$ and $\frac{3}{2}$, so that

$$P_k(y) = \frac{1}{2k+1} - \frac{2k(2k^2+1)}{3(2k+1)} \mathbb{E}\left[\frac{1}{2y-2-4X}\right]. \quad (9)$$

Proof. R_k is a rational function of y with simple poles at $y = 2j - 1$ and $R_k(\infty) = 1$. The factor with index j equals $1 - \frac{1}{y-2j+1}$. At its pole the other factors take the value $\prod_{i < j} \frac{2(j-i)-1}{2(j-i)} \prod_{i > j} \frac{2(i-j)+1}{2(i-j)} = c_{j-1}(2k - 2j + 1)c_{k-j}$, which gives the residues. Since $\sum_q (2q + 1)c_q z^q = (1 - z)^{-3/2}$ and $\sum_q c_q z^q = (1 - z)^{-1/2}$, comparing coefficients in $(1 - z)^{-3/2}(1 - z)^{-1/2} = (1 - z)^{-2}$ gives $\sum_{q=0}^{k-1} (2q + 1)c_q c_{k-1-q} = k$. The beta-binomial probability of x is $\binom{k-1}{x} (\frac{1}{2})_x (\frac{3}{2})_{k-1-x} / (2)_{k-1}$ in Pochhammer notation [7]; with $(\frac{1}{2})_x = x! c_x$ and $(\frac{3}{2})_z = (2z + 1) z! c_z$ this equals $c_x(2k - 2x - 1)c_{k-1-x}/k$. Finally $2(y - 2j + 1) = 2y - 2 - 4(j - 1)$ and $P_k = \frac{1}{2k+1} - \frac{2k^2+1}{3(2k+1)}(1 - R_k)$. $\square$

For example $w_{2,\cdot} = (3, 1)$, $w_{3,\cdot} = (\frac{15}{4}, \frac{3}{2}, \frac{3}{4})$ and $w_{4,\cdot} = (\frac{35}{8}, \frac{15}{8}, \frac{9}{8}, \frac{5}{8})$. Let ψ be the digamma function, $H_m = \sum_{i \leq m} \frac{1}{i} = \psi(m+1) - \psi(1)$ the harmonic numbers and $H_m^{\text{odd}} = \sum_{i \leq m} \frac{1}{2i-1} = \frac{1}{2}(\psi(m+\frac{1}{2}) - \psi(\frac{1}{2}))$ the odd harmonic numbers [1, 5.4.13, 5.4.15].

Proposition 6.2. *For $k \geq 1$, with $t_k = t_k(h)$,*

$$B_k = \frac{\ell_k}{2k+1} - \frac{2k^2+1}{3(2k+1)} \sum_{j=1}^k \frac{w_{k,j}}{2} \left[\psi\left(t_{k+1} - \frac{h}{2} - 2j\right) - \psi\left(t_k - \frac{h}{2} - 2j+1\right) \right].$$

The bracket equals $H_{t_{k+1}-2j-1} - H_{t_k-2j}$ for $h=0$ and $2(H_{t_{k+1}-2j-1}^{\text{odd}} - H_{t_k-2j}^{\text{odd}})$ for $h=1$. In particular B_k is rational, $B_1(0) = \frac{163}{180}$, and $S_0(10) = B_1(0) + P_2(8) + P_2(9) + P_2(10) = \frac{25493}{25200}$.

Proof. Sum Lemma 6.1 over the block, using $\sum_{n=A}^B \frac{1}{n-\frac{h}{2}-2j+1} = \psi(B-\frac{h}{2}-2j+2) - \psi(A-\frac{h}{2}-2j+1)$. $\square$

Proposition 6.3 (integral representation). *For $k \geq 1$ let*

$$W_k(x) = \sum_{j=1}^k w_{k,j} x^{4(k-j)} = \frac{4k}{\pi} \int_0^1 \theta^{1/2} (1-\theta)^{-1/2} (1-\theta+\theta x^4)^{k-1} d\theta.$$

Then

$$B_k = \frac{\ell_k}{2k+1} - \frac{2k^2+1}{3(2k+1)} \int_0^1 x^{2t_k-h-4k+1} \frac{1-x^{2\ell_k}}{1-x^2} W_k(x) dx.$$

Proof. Insert $\frac{1/2}{y-2j+1} = \int_0^1 x^{2n-h-4j+1} dx$ (with $y = n - \frac{h}{2}$) into the block sum and sum the geometric series over the block. The integral formula for W_k is $2k$ times the probability generating function of $k-1-X \sim \text{BetaBinomial}(k-1, \frac{3}{2}, \frac{1}{2})$ at x^4 , written as a binomial mixture over a $\text{Beta}(\frac{3}{2}, \frac{1}{2})$ parameter, whose normalising constant is $B(\frac{3}{2}, \frac{1}{2}) = \frac{\pi}{2}$. $\square$

Remark 6.4. For fixed k the summand $P_k(n - \frac{h}{2})$ has no rational antidifference in n : at infinity it equals $\frac{1}{2k+1} - \frac{k(2k^2+1)}{3(2k+1)n} + O(n^{-2})$, while $G(n+1) - G(n)$ has no term in $1/n$ for any rational function G . This is why the digamma function enters Proposition 6.2; in particular the ordinary rational Gosper certificate [5] does not exist.

7 Asymptotics

Let $h \in \{0, 1\}$, fix k , and write $\eta = (-1)^{k+h}$,

$$D = \frac{4}{3}k^3, \quad a_\eta = \frac{5}{3}k + \frac{1-\eta}{2}, \quad \text{so that} \quad 2t_k(h) - h - 2 = D + a_\eta$$

by (8), and $\ell_k = 2k^2 + 2k + \frac{1+\eta}{2}$. Everything below depends on h and the parity of k only through η . For $n = t_k(h) + r$ with $0 \leq r < \ell_k$, equation (9) with $y = n - \frac{h}{2}$ becomes

$$P_k\left(n - \frac{h}{2}\right) = \frac{1}{2k+1} \left[1 - \left(1 + \frac{1}{2k^2}\right) \mathbb{E}\left[\frac{1}{1 + \xi_r/D}\right] \right], \quad \xi_r = a_\eta + 2r - 4X, \quad (10)$$

because $\frac{2k(2k^2+1)}{3D} = 1 + \frac{1}{2k^2}$. Put $T_m = \sum_{r=0}^{\ell_k-1} \mathbb{E}[\xi_r^m]$. The beta-binomial law has $\mathbb{E}X = \frac{k-1}{4}$ and $\text{Var}(4X) = k^2 - 1$ [7], so

$$T_0 = \ell_k, \quad T_1 = \ell_k \mu, \quad T_2 = \ell_k \left[\mu^2 + \frac{\ell_k^2 - 1}{3} + k^2 - 1 \right], \quad \mu = 2k^2 + \frac{8}{3}k + 1.$$

In general T_m is, for each η , a polynomial in k of degree at most $2m+2$ with rational coefficients. It follows from the factorial moments $\mathbb{E}[X(X-1)\cdots(X-i+1)] = (k-1)(k-2)\cdots(k-i)(\frac{1}{2})_i/(2)_i$ and the power sums $\sum_{r < \ell_k} r^j$.

Lemma 7.1. For $k \geq 8$, $0 \leq r < \ell_k$ and $0 \leq X \leq k-1$ we have $|\xi_r/D| \leq 4/k$.

Proof. We have $-\frac{7}{3}k + 4 \leq \xi_r \leq 4k^2 + \frac{17}{3}k$, and both $(4k^2 + \frac{17}{3}k)/D = \frac{3}{k} + \frac{17}{4k^2}$ and $\frac{7}{3}k/D = \frac{7}{4k^2}$ are at most $\frac{4}{k}$. $\square$

Proposition 7.2. Let $\eta = (-1)^{k+h}$.

(a) $B_k = \frac{3}{2} - \frac{3}{4k} + \frac{3\eta}{8k^2} + O(k^{-3})$.

(b) For every $J \geq 0$ there are rational numbers $b_{j,\pm}$, independent of h , with $B_k = \sum_{j=0}^J b_{j,\eta} k^{-j} + O_J(k^{-J-1})$. Table 2 lists them for $j \leq 8$.

(c) For $k \geq 8$ and $M \geq 1$,

$$\left| B_k - \frac{1}{2k+1} \left[\ell_k - \left(1 + \frac{1}{2k^2} \right) \sum_{m=0}^M (-1)^m \frac{T_m}{D^m} \right] \right| \leq \frac{3}{2} k \frac{(4/k)^{M+1}}{1 - 4/k}.$$

Proof. Sum (10) over r and use $\frac{1}{1+u} = \sum_{m=0}^M (-u)^m + \frac{(-u)^{M+1}}{1+u}$. By Lemma 7.1 the remainder is at most $\frac{\ell_k}{2k+1} (1 + \frac{1}{2k^2}) \frac{(4/k)^{M+1}}{1-4/k}$, and $\ell_k \leq 3k^2$ gives (c). For each η , $T_m/D^m = O(k^{2-m})$ is a Laurent polynomial in k , so the bracket in (c) equals $k F_{M,\eta}(1/k)$ for a polynomial $F_{M,\eta}$ with rational coefficients, and $B_k = F_{M,\eta}(1/k)/(2 + 1/k) + O(k^{-M})$. The term with index m contributes only from the power k^{1-m} on, so the Taylor coefficients of $F_{M,\eta}(z)/(2+z)$ up to z^{M-1} do not depend on M . This gives (b).

For (a) take $M = 3$. The formulas for T_1 and T_2 and $T_3 = \sum_r \mathbb{E}[\xi_r^3] = 32k^8 + O(k^7)$ give

$$\frac{\ell_k}{2k^2} = 1 + \frac{1}{k} + O(k^{-2}), \quad \frac{T_1}{D} = 3k + 7 + \frac{25 + 3\eta}{4k} + O(k^{-2}), \quad \frac{T_2}{D^2} = 6 + \frac{21}{k} + O(k^{-2}), \quad \frac{T_3}{D^3} = \frac{27}{2k} + O(k^{-2}).$$

The bracket in (c) equals $-\frac{\ell_k}{2k^2} + (1 + \frac{1}{2k^2}) (\frac{T_1}{D} - \frac{T_2}{D^2} + \frac{T_3}{D^3}) = 3k - \frac{3(1-\eta)}{4k} + O(k^{-2})$. Dividing by $2k+1$ gives (a). $\square$

j	$b_{j,+}$	$b_{j,-}$
0	3/2	3/2
1	-3/4	-3/4
2	3/8	-3/8
3	-9/20	57/40
4	291/160	-219/160
5	-2859/560	-327/280
6	6417/560	17793/2240
7	-203421/8960	-168561/8960
8	390009/8960	274299/8960

Table 2: Coefficients of the expansion $B_k \sim \sum_j b_{j,\eta} k^{-j}$ of Proposition 7.2, with $\eta = (-1)^{k+h}$: the column $b_{j,+}$ serves even k for $h = 0$ and odd k for $h = 1$. The certificates store the coefficients for $j \leq 59$.

Put $e_k = e_k(h) = B_k - \frac{3}{2} + \frac{3}{4k}$. By Proposition 7.2(a), $e_k = (-1)^{k+h} \frac{3}{8k^2} + O(k^{-3})$, so $\sum_k e_k$ converges absolutely, and $\sum_{k>K} e_k = O(K^{-2})$.

Theorem 7.3. Let $h \in \{0, 1\}$, $N \geq 3$, and let K and τ be as in Theorem B. Then

$$S_h(N) = \kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) - \frac{1 - 3\tau(1 - \tau)(4\tau - 1)}{4K} + O(K^{-2}), \quad (11)$$

uniformly in τ , where

$$\mathcal{A}(h) = \frac{3}{2} + \frac{3}{4}\gamma + \frac{1}{4} \log \frac{3}{2} - \sum_{k=1}^{\infty} \left(B_k(h) - \frac{3}{2} + \frac{3}{4k} \right). \quad (12)$$

At the block ends $N_K = t_{K+1}(h) - 1$ this reads $S_h(N_K) = \kappa N_K^{1/3} - \frac{1}{4} \log N_K - \mathcal{A}(h) - \frac{1}{4K} + O(K^{-2})$.

Proof. Block ends. By Theorem 5.4, $S_h(N_K) = \sum_{k \leq K} B_k = \frac{3}{2}K - \frac{3}{4}H_K + \sum_{k \leq K} e_k$, where $H_K = \log K + \gamma + \frac{1}{2K} + O(K^{-2})$. Hence

$$S_h(N_K) = \frac{3}{2}K - \frac{3}{4} \log K - \frac{3}{4}\gamma - \frac{3}{8K} + \sum_{k \geq 1} e_k + O(K^{-2}).$$

By (8), $N_K = \frac{2}{3}(K+1)^3 + \frac{5}{6}(K+1) + O(1)$, so $\kappa N_K^{1/3} = \frac{3}{2}(K+1) + \frac{5}{8K} + O(K^{-2})$ and $\log N_K = \log \frac{2}{3} + 3 \log K + \frac{3}{K} + O(K^{-2})$. Combining these expansions with (12) gives the block-end formula, because $-\frac{3}{8} - \frac{5}{8} + \frac{3}{4} = -\frac{1}{4}$.

General N . The case $K = 1$ is a finite statement, so let $K \geq 2$. Put $m = t_K(h) - 1 = N_{K-1}$, $r = N - m$ and $D_K = t_{K+1}(h) - t_K(h) = 2K^2 + 2K + O(1)$, so that $\tau = r/D_K$. The block-end formula for $K - 1$ gives $S_h(m) = \kappa m^{1/3} - \frac{1}{4} \log m - \mathcal{A}(h) - \frac{1}{4K} + O(K^{-2})$. For $1 \leq s \leq r$, equation (10) with $\xi = a_\eta + 2s - 2 - 4X$ applies to $n = t_K(h) - 1 + s$. Here $\mathbb{E}\xi = \frac{2}{3}K + 2s - \frac{1+\eta}{2}$ and $\mathbb{E}\xi^2 = (\mathbb{E}\xi)^2 + K^2 - 1$, and expanding $\frac{1}{1+u}$ to second order with Lemma 7.1 gives, uniformly in $1 \leq s < D_K$,

$$P_K\left(t_K(h) - 1 + s - \frac{h}{2}\right) = \frac{3s}{4K^4} - \frac{3s}{8K^5} - \frac{9s^2}{8K^7} + O(K^{-4}).$$

Summing over $s \leq r$ and inserting $r = \tau D_K$,

$$S_h(N) - S_h(m) = \frac{3}{2}\tau^2 + \frac{\frac{9}{4}\tau^2 - 3\tau^3}{K} + O(K^{-2}).$$

Since $r/m = O(1/K)$, Taylor expansion gives

$$\kappa N^{1/3} - \kappa m^{1/3} = \frac{3}{2}\tau + \frac{3(\tau - \tau^2)}{2K} + O(K^{-2}), \quad \frac{1}{4} \log \frac{N}{m} = \frac{3\tau}{4K} + O(K^{-2}).$$

Adding up, the coefficient of $1/K$ is $-\frac{1}{4} - \frac{3}{4}\tau + \frac{15}{4}\tau^2 - 3\tau^3 = -\frac{1}{4}(1 - 3\tau(1 - \tau)(4\tau - 1))$, and the terms of order one give $-\frac{3}{2}\tau(1 - \tau)$. $\square$

Theorem B follows, because $K \asymp N^{1/3}$.

8 The Alper constant

Definition 8.1. For $h \in \{0, 1\}$ let $N_K = t_{K+1}(h) - 1$ be the zeros of P , that is, the ends of the blocks. Put

$$\mathcal{A}(h) = \lim_{K \rightarrow \infty} \left(\kappa N_K^{1/3} - \frac{1}{4} \log N_K - S_h(N_K) \right).$$

The Alper constant is $\mathcal{A} = \mathcal{A}(0)$.

By Theorem 7.3 the limit exists and is given by the series (12); Section 10 extends the definition to every $h \notin \mathcal{E}$. In the same way as Euler's constant measures how far the harmonic numbers lie from $\log N$, the Alper constant measures how far the equilibrium sums $S_0(N)$ lie from their smooth approximation $\kappa N^{1/3} - \frac{1}{4} \log N$; along general N the parabola $\frac{3}{2}\tau(1 - \tau)$ of Theorem B has to be added.

Representations

The series (12) has rational terms $e_k = B_k - \frac{3}{2} + \frac{3}{4k}$, which Proposition 6.2 evaluates exactly, and it converges absolutely. Its leading terms alternate and sum in closed form: since $\sum_k (-1)^{k+1} k^{-2} = \frac{\pi^2}{12}$, we have $\sum_k (-1)^{k+h} \frac{3}{8k^2} = (-1)^{h+1} \frac{\pi^2}{32}$. Thus

$$\sum_{k=1}^{\infty} e_k(h) = (-1)^{h+1} \frac{\pi^2}{32} + \sum_{k=1}^{\infty} \left(e_k(h) - (-1)^{k+h} \frac{3}{8k^2} \right),$$

and the last series, with terms $O(k^{-3})$, equals

$$\begin{aligned} &0.5655231753318100092599469738976772679775991822265765812903645817609304 \dots \quad (h = 0), \\ &0.0794352347285882502563253866959460981430671626443740113590515840927113 \dots \quad (h = 1). \end{aligned}$$

Subtracting further terms of Proposition 7.2 produces the contributions $\frac{381}{320}\zeta(3)$ and $-\frac{1497}{1280}\zeta(4)$ for $h = 0$, and $-\frac{69}{320}\zeta(3)$ and $\frac{2073}{1280}\zeta(4)$ for $h = 1$. Each such step is an exact identity and accelerates convergence, but we do not claim that the expansion in $1/k$ converges. Its coefficients in Table 2 grow quickly. Finally, Proposition 6.3 turns (12) into a series of explicit integrals.

Certified digits

Theorem 8.2. *The values of $\mathcal{A} = \mathcal{A}(0)$ and of $\mathcal{A}(1)$ printed in the introduction are correctly rounded to 80 decimal places. Moreover*

$$\begin{aligned} \sum_{k \geq 1} e_k(0) &\approx 0.2570980377977675524213691301515475449990 \\ &4607575030187421494741375405402095091354 \end{aligned}$$

is correctly rounded to 80 decimal places.

Method. For each h , split $\sum_k e_k$ at $K_0 = 192$. The 191 head terms come from Proposition 6.2. In the tail, B_k is replaced by the expansion of Proposition 7.2 through $j = 59$, with the exact rational coefficients computed from the polynomials T_m . The tail sums are Hurwitz zeta values: if a_0 and a_1 are the least even and odd integers $\geq K_0$, then $\sum_{k \geq K_0, k \equiv a_i(2)} k^{-j} = 2^{-j} \zeta(j, a_i/2)$. Two errors remain. Proposition 7.2(c) with $M = 60$ bounds the geometric truncation in each block. The Taylor truncation leaves $G(z)/(2+z)$, where $G = F_{60,\eta} - (2+z) \sum_{j \leq 59} b_{j,\eta} z^j$ is an explicit polynomial divisible by z^{60} , so it is at most $\frac{1}{2} \sum_p |g_p| k^{-p}$ in terms of the coefficients g_p of G . Both errors are summed over $k \geq K_0$ with $\sum_{k \geq K_0} k^{-s} \leq K_0^{-s} + K_0^{1-s}/(s-1)$. The two bounds are below $3.481 \cdot 10^{-100}$ and $1.26 \cdot 10^{-109}$ for both values of h .

All transcendental quantities (ψ , Hurwitz ζ , γ and $\log \frac{3}{2}$) are evaluated in Arb ball arithmetic [6] at 160 digits, via python-flint [4]. The resulting enclosures have radius below $7.4 \cdot 10^{-100}$, and the program checks that the rounding to 80 places is constant on each ball. For $h = 0$ a second run with $K_0 = 128$, the expansion through $j = 47$ and 110 digits gives an enclosure of radius below $6 \cdot 10^{-71}$ that overlaps the first. For both h , an independent 130-digit mpmath summation differs from the certified midpoint by less than 10^{-93} . This is a computer-assisted evaluation of the error bounds above, not a formal proof. $\square$

Is there a closed form?

We searched for an integer relation between $\mathcal{A}$ and

$$\begin{aligned} &1, \gamma, \log 2, \log 3, \pi, \pi^2, \pi^4, \zeta(3), \zeta(5), G, \\ &\log \pi, \log \Gamma(\tfrac{1}{4}), \log \Gamma(\tfrac{1}{3}), \sqrt{2}, \pi\sqrt{3}, \log^2 2, \log^2 3, \log 2 \log 3, \pi \log 2, \end{aligned}$$

where G is Catalan's constant, using PSLQ [3] in 120-digit arithmetic with tolerance 10^{-85} and coefficients of absolute value at most 1000. The same search was run for $\mathcal{A}(1)$, for $\mathcal{A}(h) - \frac{1}{4} \log \frac{3}{2}$ and for $\sum_k e_k(h)$, and also against the subset $1, \gamma, \log 2, \log 3, \pi^2, \zeta(3)$. No relation was found. These finite searches do not establish the absence of a closed form.

Two facts make $\mathcal{A}$ less arbitrary than an empirical constant. First, the block ends $t_k(0)$ are exact polynomials in k for each parity, by (8); the fractional positions of the zeros β_k never enter. Second, every ingredient of (12) is explicit: rational block sums and the constants γ and $\log \frac{3}{2}$.

How the constants were found

The constant $\mathcal{A}(1)$ was seen first, numerically. Exact block sums give $S_1(10^m)$ to many digits for $m \leq 20$. The defect $\kappa N^{1/3} - \frac{1}{4} \log N - S_1(N)$ oscillates inside each block. Subtracting $\frac{3}{2}\tau(1-\tau)$ from this defect leaves values clustering near 1.6464. The proofs did not use the special payments $2m-1$ except in the explicit form of the thresholds, which led to the family $G_n(h)$ and to its simplest member $h=0$, whose constant is the Alper constant. Theorem 7.3 and the certified evaluations turned these observations into theorems.

9 Numerical data

Table 3 lists $S_0(10^m)$ for $1 \leq m \leq 20$, computed in two ways. The certified computation uses Arb balls: for $m \leq 9$ it adds all block sums of Proposition 6.2, and for $m \geq 10$ it uses $S(N) = \frac{3}{2}(K-1) - \frac{3}{4}H_{K-1} + \sum_k e_k - \sum_{k \geq K} e_k$ plus the partial block, with the tail and the partial block taken from Proposition 7.2. It certifies 30 decimals. The second computation adds all block sums in 100-digit decimal arithmetic, switching to a bounded moment expansion of each block for $k \geq 1000$. The two agree to within $5 \cdot 10^{-31}$. The fourth column shows the normalised residual $K \delta(N)$ of the first-order formula, where

$$\delta(N) = S_0(N) - \left(\kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A} - \frac{3}{2}\tau(1-\tau) \right),$$

and the last column shows its limit $c(\tau) = -\frac{1}{4}(1 - 3\tau(1-\tau)(4\tau-1))$ predicted by Theorem 7.3. The difference between the two columns is $O(1/K)$, and it is below 10^{-5} for $m \geq 15$.

N	$S_0(N)$	$S_0(N)/N^{1/3}$	$K \delta(N)$	$c(\tau)$
10^1	1.01162698412698412698	0.469555651331	-0.164628	-0.268039
10^2	4.76382820302819364808	1.026335673796	-0.203027	-0.275258
10^3	13.30208678375677871897	1.330208678376	-0.089062	-0.151146
10^4	32.56774386666910234077	1.511660762675	0.041852	0.017932
10^5	74.87679399375861732781	1.613171624585	-0.287017	-0.291100
10^6	166.10210752872543471053	1.661021075287	-0.085058	-0.089064
10^7	363.77124561032651935744	1.688476551614	0.013286	0.010878
10^8	790.28053769454961122398	1.702607805266	-0.197141	-0.198360
10^9	1709.80689936028358017628	1.709806899360	0.034865	0.034247
10^{10}	3691.53391596882046613243	1.713458260326	-0.269046	-0.269115
10^{11}	7961.51938003709654146938	1.715257353771	-0.223351	-0.223461
10^{12}	17161.84551749443685308992	1.716184551749	-0.289401	-0.289424
10^{13}	36983.76129919255133262475	1.716634134713	-0.291166	-0.291168
10^{14}	79689.45681287532869805299	1.716857301875	-0.114766	-0.114791
10^{15}	171696.35811503857501272212	1.716963581150	-0.122306	-0.122312
10^{16}	369920.57702331276152428602	1.717019219635	-0.270994	-0.270996
10^{17}	796982.05832200729109138505	1.717045793782	-0.228871	-0.228872
10^{18}	1717058.94943545492359063880	1.717058949435	-0.254104	-0.254105
10^{19}	3699305.29382848385618983787	1.717065414396	-0.286261	-0.286261
10^{20}	7969925.78324778285942282425	1.717068458441	-0.016978	-0.016979

Table 3: $S_0(N)$ rounded to 20 decimals, the ratio $S_0(N)/N^{1/3}$, and the second-order residual of Theorem 7.3. The limit of the ratio is $\kappa = 1.7170713\dots$

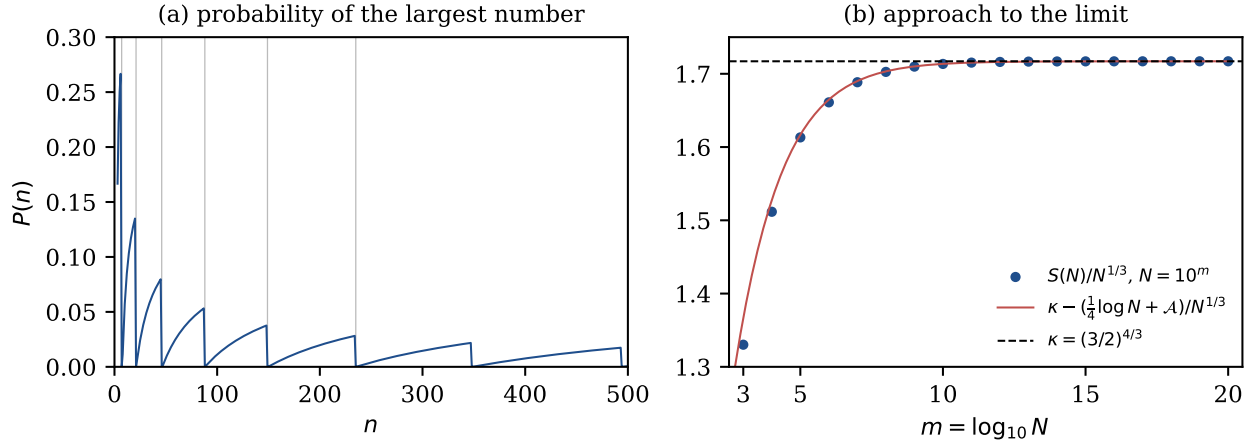


Figure 1: The game $G_n(0)$. (a) The probability $P(n)$ of the largest number for $3 \leq n \leq 500$; the grey lines mark the zeros $n = t_k(0) - 1$. Within the positive part of each block P increases. It drops to 0 one step before the support widens. (b) $S_0(N)/N^{1/3}$ at $N = 10^m$ for $3 \leq m \leq 20$ (dots), the limit κ (dashed), and the curve $\kappa - (\frac{1}{4} \log N + \mathcal{A})/N^{1/3}$.

10 The Alper function

The proofs of Sections 6 and 7 used $h \in \{0, 1\}$ only through the explicit thresholds (8). For other $h \notin \mathcal{E}$ the thresholds differ from those of $h = 0$ or $h = 1$ only for finitely many k , and the arguments carry over. The constant $\mathcal{A}(h)$ thereby becomes a function of h , whose value at 0 is the Alper constant.

Throughout this section

$$h_0 = \lfloor h \rfloor \in \{0, 1\}, \quad \varphi = h - h_0 \in [0, 1).$$

Lemma 10.1. *For every $k \geq 2$, $\vartheta_k < \frac{1}{2k}$. Consequently, if $h \in [0, 2) \setminus \mathcal{E}$ and $k \geq 2$, then $t_k(h) = t_k(h_0) + 1$ when $k \equiv h_0 \pmod{2}$ and $2\vartheta_k > 1 - \varphi$, and $t_k(h) = t_k(h_0)$ otherwise. In particular $t_k(h) = t_k(h_0)$ for every $k \geq 1/(1 - \varphi)$.*

Proof. Retain U , G , $d = k - \frac{1}{2}$, $v = \frac{4k^2-3}{12}$ and $x_0 = \frac{k(4k^2-1)}{6}$ from the proof of Lemma 5.2, and $V = [-\frac{k}{2}, \frac{k}{2}]$ from Remark 5.5. Let $z = x_0 + \frac{1}{2k}$. As in the proof of Lemma 5.2, $G(z) \leq \frac{k}{z}(1 + \frac{v}{z^2-d^2})$, while $\lambda_k = \int_V \frac{x_0 dt}{x_0^2-t^2} \geq \frac{k}{x_0}(1 + \frac{k^2}{12x_0^2})$. Since $z > x_0$, the inequality $G(z) < \lambda_k$ follows from $\frac{x_0 v}{x_0^2-d^2} < \frac{1}{2k} + \frac{k^2}{12x_0}$, which, as $v - \frac{k^2}{12} = \frac{k^2-1}{4}$, is equivalent to

$$2k\left(\frac{k^2-1}{4}x_0^2 + \frac{k^2 d^2}{12}\right) < x_0(x_0^2 - d^2).$$

Here $2k \cdot \frac{k^2-1}{4}x_0^2 = \frac{3(k^2-1)}{4k^2-1}x_0^3 < \frac{3}{4}x_0^3$, so it suffices that $\frac{k^3 d^2}{6} + x_0 d^2 \leq \frac{x_0^3}{4}$. This holds because $d < k$ and $x_0 \geq \frac{5}{8}k^3$ give $\frac{k^3 d^2}{6} + x_0 d^2 < \frac{19}{15}x_0 k^2 \leq \frac{25}{256}k^6 x_0 \leq \frac{x_0^3}{4}$ for $k \geq 2$. Hence $G(z) < \lambda_k$, so $x^* < z$, that is, $\vartheta_k < \frac{1}{2k}$.

The description of $t_k(h)$ is Remark 5.5 written in terms of h_0 and φ , using $t_k(1) = t_k(0) + 1$ for even k : for $h_0 = 0$ exactly the even k with $h > 1 - 2\vartheta_k$ are raised by one, and for $h_0 = 1$ exactly the odd k with $h > 2 - 2\vartheta_k$. In both cases this means $2\vartheta_k > 1 - \varphi$, which is impossible for $k \geq 1/(1 - \varphi)$, because $2\vartheta_k < \frac{1}{k}$. $\square$

Definition 10.2. For $h \in [0, 2) \setminus \mathcal{E}$ let $\mathcal{A}(h)$ be the limit of Definition 8.1, taken along the ends $N_K = t_{K+1}(h) - 1$ of the blocks of $G_n(h)$, which are exactly the zeros of P . The function $h \mapsto \mathcal{A}(h)$ is the *Alper function*.

Theorem 10.3. *Let $h \in [0, 2) \setminus \mathcal{E}$, and let N , K and τ be as in Theorem B. The limit defining $\mathcal{A}(h)$ exists and is given by the series (12), with the block sums $B_k(h) = \sum_{n=t_k(h)}^{t_{k+1}(h)-2} P_k(n - \frac{h}{2})$, and*

$$S_h(N) = \kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) - \frac{1 - 3\varphi - 3\tau(1 - \tau)(4\tau - 1)}{4K} + O_h(K^{-2}), \quad (13)$$

uniformly in τ . Moreover, for $k \geq \max(8, 1/(1 - \varphi))$ and $\eta = (-1)^{k+h_0}$,

$$B_k(h) = \frac{3}{2} - \frac{3}{4k} + \left(\frac{3\eta}{8} - \frac{3\varphi}{4}\right) \frac{1}{k^2} + O(k^{-3}),$$

with an implied constant that does not depend on h .

Proof. Let $k \geq \max(8, 1/(1 - \varphi))$. By Lemma 10.1, $t_k(h)$ and $t_{k+1}(h)$ agree with the thresholds of h_0 . So the block k has the length ℓ_k of Section 6 for h_0 , and $2t_k(h) - h - 2 = D + a_\eta - \varphi$. Thus (10) holds with $\xi_r = a_\eta - \varphi + 2r - 4X$. Lemma 7.1 remains valid, because the shift lowers the lower bound of ξ_r only to $-\frac{7}{3}k + 3$, and so does Proposition 7.2(c), with the moments T_m of the shifted ξ_r . The shift lowers T_1/D by $\varphi \ell_k/D = \frac{3\varphi}{2k} + O(k^{-2})$ and changes T_2/D^2 and T_3/D^3 by $O(k^{-2})$. In the proof of Proposition 7.2(a) the bracket therefore becomes $3k - (\frac{3(1-\eta)}{4} + \frac{3\varphi}{2})\frac{1}{k} + O(k^{-2})$, and dividing by $2k + 1$ gives the expansion of $B_k(h)$, uniformly in $\varphi \in [0, 1)$.

Hence $e_k(h) = B_k(h) - \frac{3}{2} + \frac{3}{4k} = O(k^{-2})$, the series (12) converges absolutely, and, since $\sum_{k>K} (-1)^k k^{-2} = O(K^{-2})$ and $\sum_{k>K} k^{-2} = \frac{1}{K} + O(K^{-2})$,

$$\sum_{k>K} e_k(h) = -\frac{3\varphi}{4K} + O(K^{-2}).$$

For large K the block ends $N_K = t_{K+1}(h) - 1$ are those of h_0 , so the block-end part of the proof of Theorem 7.3 applies verbatim, except that the tail now contributes $\frac{3\varphi}{4K}$ to $S_h(N_K)$. This gives $S_h(N_K) = \kappa N_K^{1/3} - \frac{1}{4} \log N_K - \mathcal{A}(h) - \frac{1-3\varphi}{4K} + O(K^{-2})$, which is (13) at the block ends, and it shows that the limit exists and equals (12). In the part for general N the shift changes $\mathbb{E}\xi$ by less than 1, and hence P_K only by $O(K^{-4})$, so the expansion of P_K used there, and with it the rest of the argument, is unchanged. $\square$

Proposition 10.4. *Let $\mathcal{A}$ be the Alper function on $[0, 2) \setminus \mathcal{E}$.*

(a) *On every interval $J \subset [0, 2)$ without points of $\mathcal{E}$ the thresholds $t_k(h)$ do not depend on h , and $\mathcal{A}$ is continuous, strictly increasing and strictly convex on J .*

(b) *Let $h^* \in \mathcal{E}$. The one-sided limits $\mathcal{A}(h^* \pm)$ exist, and*

$$\mathcal{A}(h^*+) - \mathcal{A}(h^*-) = - \sum_k P_{k-1}(\beta_k - 1) < 0,$$

where k runs over the finitely many $k \geq 2$ with $\beta_k + \frac{h^}{2} \in \mathbb{Z}$.*

(c) *$\mathcal{A}(h) \rightarrow \mathcal{A}(1)$ as $h \rightarrow 1^-$, and $\mathcal{A}(h) \rightarrow \mathcal{A}(0)$ as $h \rightarrow 2^-$.*

Proof. (a) A threshold $t_k(h) = \lfloor \beta_k + \frac{h}{2} \rfloor + 1$ can change only where $\beta_k + \frac{h}{2} \in \mathbb{Z}$, that is, at a point of $\mathcal{E}$. Since $\mathcal{E}$ accumulates at 1 and 2 from below, J lies in $[0, 1)$ or in $[1, 2)$, and $\varphi \leq \varphi_J$ on J for some $\varphi_J < 1$. By Theorem 10.3, $|e_k(h)| \leq Ck^{-2}$ on J for $k \geq \max(8, 1/(1 - \varphi_J))$, so the series (12) converges uniformly on J , and $\mathcal{A}$ is continuous there. By (9), $P_k(y) = \frac{1}{2k+1} - c \mathbb{E}[(2y - 2 - 4X)^{-1}]$ with $c > 0$, and $2y - 2 - 4X \geq 2y - 4k + 2 > 0$ in the block k , where $y > \beta_k > 2k$ (for $k = 1$, $y > 2$). Hence every term $P_k(n - \frac{h}{2})$, and with it every $e_k(h)$, is strictly decreasing and strictly concave in $h \in J$. So is $\sum_k e_k(h)$, and $\mathcal{A}$ is strictly increasing and strictly convex.

(b) The point h^* is isolated in $\mathcal{E}$, so (a) applies on intervals $(h^* - \varepsilon, h^*)$ and $(h^*, h^* + \varepsilon)$, with thresholds t_k^- and t_k^+ . The uniform convergence in (a) gives $\mathcal{A}(h^* \pm) = \frac{3}{2} + \frac{3}{4}\gamma + \frac{1}{4} \log \frac{3}{2} - \sum_k e_k^\pm(h^*)$, where $e_k^\pm$ is computed with the thresholds $t_k^\pm$. These differ only for the k with $m = \beta_k + \frac{h^*}{2} \in \mathbb{Z}$, where $t_k^- = m$ and $t_k^+ = m + 1$: the block $k - 1$ gains the number $m - 1$, with $P_{k-1}(m - 1 - \frac{h^*}{2}) = P_{k-1}(\beta_k - 1)$, and the block k loses the number m , with $P_k(\beta_k) = 0$. There are finitely many such k by Lemma 10.1, and $P_{k-1}(\beta_k - 1) > 0$ by Remark 5.5.

(c) Let $\tilde{B}_k(h)$ be the sum of $P_k(n - h/2)$ over $t_k(h_0) \leq n \leq t_{k+1}(h_0) - 2$, even when these are not the actual thresholds for h . The algebra in the proof of Theorem 10.3 applies to this sum for every $k \geq 8$ and gives $\tilde{B}_k(h) - 3/2 + 3/(4k) = O(k^{-2})$, uniformly in h . By Lemma 10.1, the actual block differs by at most its first and its last number. To include the possible extra last number, take $0 \leq r \leq \ell_k$ in $\xi_r = a_\eta - \varphi + 2r - 4X$, with ℓ_k and a_η for h_0 . Then

$$-\frac{7}{3}k + 3 \leq \xi_r \leq 4k^2 + \frac{17}{3}k + 2, \quad |\xi_r/D| \leq 4/k \quad (k \geq 8).$$

Equation (10) therefore gives the uniform bound

$$\left| P_k \left(t_k(h_0) + r - \frac{h}{2} \right) \right| \leq \frac{4/k + 1/(2k^2)}{(2k+1)(1-4/k)} = O(k^{-2}).$$

Consequently $|e_k(h)| \leq Ck^{-2}$ for all $k \geq 8$ and all $h \in [0, 2) \setminus \mathcal{E}$, with one constant C . Each of the finitely many earlier terms is bounded as well, since its denominators stay positive for $0 \leq h \leq 2$. Dominated convergence now reduces the endpoint limits to those of each $e_k(h)$. As $h \rightarrow 1^-$, eventually $t_k(h) = t_k(1)$ and $t_{k+1}(h) = t_{k+1}(1)$, so $e_k(h) \rightarrow e_k(1)$. As $h \rightarrow 2^-$, eventually $t_k(h) = t_k(0) + 1$ for each fixed $k \geq 2$, and $P_k(n - \frac{h}{2}) \rightarrow P_k(n - 1)$; hence $B_k(h) \rightarrow B_k(0)$ for $k \geq 2$, and $B_1(h) \rightarrow P_1(2) + B_1(0) = B_1(0)$. $\square$

For $h \geq 2$ the rules of $G_n(h)$ still make sense, but some payments are ≤ 0 . We keep the notation $P(n)$ and $S_h(N)$ as long as all equilibria of the games involved assign the same probability to the largest number, and we define $\mathcal{A}(h)$ as the limit of $\kappa N^{1/3} - \frac{1}{4} \log N - S_h(N)$ along the zeros N of P , when it exists.

Proposition 10.5. *Let $h \in [0, 2) \setminus \mathcal{E}$ and $j \geq 1$.*

- (a) *For $n \geq j + 3$ the game $G_n(h + 2j)$ has a unique equilibrium, obtained from that of $G_{n-j}(h)$ by raising every number by j . In particular, $P(n)$ for $G_n(h + 2j)$ equals $P(n - j)$ for $G_{n-j}(h)$.*
- (b) *Every equilibrium of $G_3(h + 2)$ assigns probability 0 to the number 3. Hence $S_{h+2}(N) = S_h(N - 1)$ for $N \geq 3$, with $S_h(2) = 0$, and $\mathcal{A}(h + 2) = \mathcal{A}(h)$.*

Proof. (a) Lowering every number by j preserves all differences, and a win with the lowered number m pays $2(m + j) - (h + 2j) = 2m - h$. Thus $G_n(h + 2j)$ is the game on $\{1 - j, \dots, n - j\}$ with the payments $2m - h$, and its numbers $1, \dots, n - j$ form $G_{n-j}(h)$. Let p be the equilibrium of $G_{n-j}(h)$, extended by zero. Against p , a number $m \leq 0$ receives $2m - h \leq 0$ from each $i \geq 1$ with $i - m$ odd and pays $2i - h > 0$ to each $i \geq 1$ with $i - m$ even. The support of p consists of at least three consecutive numbers, so it meets both parities, and $f_m < 0$. Thus p satisfies (2). If q is any equilibrium, then

$$\sum_i q_i f_i(p) = q^T A p = -p^T A q = -\sum_i p_i f_i(q) \geq 0,$$

while each term $q_i f_i(p)$ is ≤ 0 . Hence $q_i = 0$ whenever $f_i(p) < 0$, in particular for $i \leq 0$, and q is an equilibrium of $G_{n-j}(h)$; by Theorem 5.4, $q = p$.

(b) After lowering by one, $G_3(h + 2)$ is the game on $\{0, 1, 2\}$. The pure strategy 1 is an equilibrium, since against it $f_0 = -h \leq 0$ and $f_2 = -(2 - h) < 0$. As in (a), every equilibrium assigns probability 0 to the number 2, which is the original 3. Together with (a) for $j = 1$ this gives $S_{h+2}(N) = S_h(N - 1)$. The zeros of P for $h + 2$ are 3 and the numbers $N_K + 1$, and $\kappa(N_K + 1)^{1/3} - \frac{1}{4} \log(N_K + 1) - S_h(N_K) \rightarrow \mathcal{A}(h)$. $\square$

Remark 10.6. The identity $\mathcal{A}(h + 2) = \mathcal{A}(h)$ in Proposition 10.5 has $0 \leq h < 2$, $h \notin \mathcal{E}$, as its base range; it does not assert global periodicity. For $j \geq 2$ and $N \geq j + 2$, Proposition 10.5(a) gives $S_{h+2j}(N) = c_j(h) + S_h(N - j)$, where $c_j(h)$ is the sum of the values $P(n)$ of the first games $G_n(h + 2j)$, $3 \leq n \leq j + 2$, provided these are defined; then $\mathcal{A}(h + 2j) = \mathcal{A}(h) - c_j(h)$. For $h \in (0, 2) \setminus \mathcal{E}$ the argument of the proof above shows that “always 3” is the unique equilibrium of $G_3(h + 4)$ and of $G_4(h + 4)$, so $c_2(h) = 1$ and $\mathcal{A}(h + 4) = \mathcal{A}(h) - 1$. For $j \geq 3$ the correction must in general be determined from these finitely many initial games.

At a positive odd integer h every payment is odd. Then every principal submatrix of A of even order has an odd Pfaffian and is nonsingular, and every $G_n(h)$ has a unique equilibrium: if $p \neq q$ were equilibria and U the union of their supports, the argument of the proof above would give $A_U p_U = A_U q_U = 0$, although the rank of A_U is at least $|U| - 1$. This extends the argument modulo 2 mentioned in Section 3, and $\mathcal{A}$ is defined at every positive odd integer. The first games give $c_3(1) = \frac{14}{9}$ and $c_4(1) = \frac{91}{45}$, so

$$\mathcal{A}(3) = \mathcal{A}(1), \quad \mathcal{A}(5) = \mathcal{A}(1) - 1, \quad \mathcal{A}(7) = \mathcal{A}(1) - \frac{14}{9}, \quad \mathcal{A}(9) = \mathcal{A}(1) - \frac{91}{45}.$$

At an even integer $h = 2j \geq 4$ a win with j pays 0, and “always j ” and “always $j + 1$ ” are both equilibria of $G_{j+1}(2j)$. So $P(j + 1)$ is not defined, and neither is $\mathcal{A}(2j)$, while $\mathcal{A}(2) = \mathcal{A}(0)$.

Values

Table 4 lists $\mathcal{A}(h)$ for $h = 0, 0.1, \dots, 1.9$, and Figure 2 shows the graph of $\mathcal{A}$ on $[0, 2)$. The values are computed as in Theorem 8.2: the thresholds are found by exact sign tests for $k \leq 193$ and are regular beyond, by Lemma 10.1, and the tail uses the exact expansion of the regular blocks with the offset $a_\eta - \varphi$. The Arb enclosures have radius below $3 \cdot 10^{-54}$, and the values at $h = 0$ and $h = 1$ agree with Theorem 8.2. The first points of $\mathcal{E}$ are

$$\begin{aligned} 6 - 2\sqrt{7} &= 0.70849\dots, & 1 - 2\vartheta_4 &= 0.82150\dots, & 1 - 2\vartheta_6 &= 0.87762\dots, \\ 2 - 2\vartheta_3 &= 1.77198\dots, & 2 - 2\vartheta_5 &= 1.85456\dots, \end{aligned}$$

where $6 - 2\sqrt{7}$ comes from $\beta_2 = 5 + \sqrt{7}$. It is the first exceptional point: for even $k \geq 4$, Lemma 10.1 gives $\vartheta_k < 1/8 < \vartheta_2$, while odd k give points above 1. At this first point $\mathcal{A}$ falls from $2.03182\dots$ to $1.75753\dots$, a jump of $P_1(4 + \sqrt{7}) = \frac{\sqrt{7}-1}{6} = 0.27429\dots$. On a grid of 3207 points the largest value of $\mathcal{A}$ lies just below $6 - 2\sqrt{7}$, and the smallest is $\mathcal{A}(1)$.

h	$\mathcal{A}(h)$	h	$\mathcal{A}(h)$
0.0	1.7771799879	1.0	1.6464176534
0.1	1.8116524222	1.1	1.6854092798
0.2	1.8465611039	1.2	1.7252229864
0.3	1.8819309085	1.3	1.7659309340
0.4	1.9177891039	1.4	1.8076154543
0.5	1.9541656688	1.5	1.8503710032
0.6	1.9910936650	1.6	1.8943065925
0.7	2.0286096750	1.7	1.9395488435
0.8	1.7913637669	1.8	1.8508040685
0.9	1.7105887967	1.9	1.8174408673

Table 4: The Alper function, rounded to 10 decimals. The values at 0.8, 0.9, 1.8 and 1.9 lie beyond the first points of $\mathcal{E}$, where $\mathcal{A}$ jumps down.

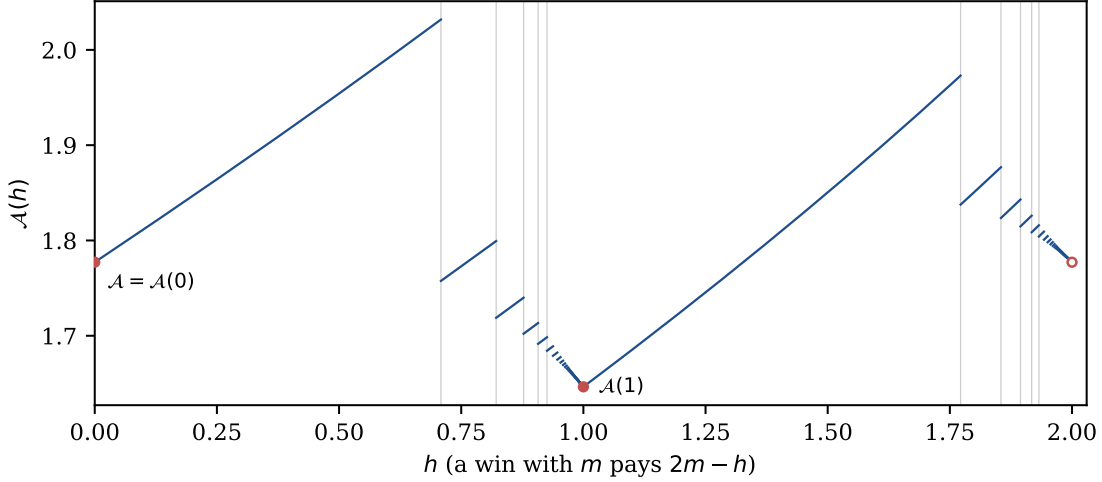


Figure 2: The Alper function on $[0, 2]$, drawn on each interval between consecutive points of $\mathcal{E}$ with $k \leq 120$; grey lines mark the first points of $\mathcal{E}$. The filled dots are $\mathcal{A} = \mathcal{A}(0)$ and $\mathcal{A}(1)$. The open dot at $h = 2$ is the limit $\mathcal{A}(0)$ of $\mathcal{A}(h)$ as $h \rightarrow 2^-$, which equals $\mathcal{A}(2)$ by Proposition 10.5.

11 Concluding remarks

The structure of the equilibria rests on two features of the payments: they grow linearly, which makes the payoff differences in Lemma 3.1 positive and the neighbour equations solvable, and the largest payment enters only through $y = n - \frac{h}{2}$. In the asymptotics the parameter h acts in two ways: through the finitely many irregular thresholds, which cause the jumps of the Alper function, and through the shift φ of the regular blocks, which appears in the term of order $1/K$.

We list some questions that we have not answered.

1. Does $\mathcal{A}$ have a closed form in classical constants? Is it irrational?
2. Is ϑ_k strictly decreasing? Then every point of $\mathcal{E}$ comes from a single k , and the sum in Proposition 10.4(b) has one term; this holds for $k \leq 400$. More generally, how does $\mathcal{A}(h)$ approach $\mathcal{A}(1)$ and $\mathcal{A}(0)$ as $h \rightarrow 1^-$ and $h \rightarrow 2^-$, where the jumps and the gaps between the points of $\mathcal{E}$ are both of order k^{-2} ?
3. What is the nature of the series $\sum_j b_{j,\pm} k^{-j}$ in Table 2? Its coefficients grow quickly, and we expect it to diverge.
4. The method of Theorem 7.3 should give further terms of $S_h(N)$ in powers of $1/K$, with polynomial coefficients in τ . A general description of these polynomials would be interesting.

The exact checks of all displayed identities and examples, the computation of the tables and the figures, and the programs that certify $\mathcal{A}$, $\mathcal{A}(1)$ and the values of the Alper function are included in the source archive. Project repository: <https://github.com/AlperTheKing/alper-constant>. AI assistance was used in the derivations, in writing and running the verification code, and in preparing this manuscript. Review status: Author reviewed. The results have not been refereed or verified in a proof assistant. This is an unrefereed preprint.

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